
MACRO FOUNDATIONS — WHAT MOVES THE WHOLE MARKET

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Abstract

Individual stocks rise and fall for a thousand idiosyncratic reasons, but the *whole market* — every index, every risk asset, at once — is moved by a much smaller set of forces. This paper builds those forces from zero for a reader with no economics background: *growth*, *inflation*, *interest rates*, and *liquidity*, the four master variables, each defined plainly and shown in one reproducible chart. From there it treats the business cycle and how it is dated; inflation from first principles, with the 2021–23 episode as the modern teaching case; the central bank's policy rate and the yield curve read as a picture; quantitative easing and tightening in plain mechanics; the handful of scheduled data releases that reliably move prices, measured on free data; and the loose, regime-dependent way shocks transmit across rates, equities, credit, currencies and commodities. A deliberate through-line is calibration: this is a self-taught author's learning vehicle, so every causal claim is stated with how firmly the evidence supports it, and every "X moves Y" is paired with a period in which it did not. The paper closes on what macro cannot do — it is excellent for context and poor for timing — a glossary of about sixty terms, and a reading ladder. Every number is computed from freely available public data and is reproducible from the accompanying code.

Keywords — Macroeconomics, growth, GDP, business cycle, inflation, CPI, PCE, interest rates, federal funds rate, yield curve, inversion, central bank, quantitative easing, liquidity, financial conditions, data releases, FOMC, cross-asset transmission, risk-on risk-off, regimes

1 Introduction

1.1 Motivation and thesis

Open any trading platform on a day the *Consumer Price Index* is released and watch the whole board move together: equity-index futures, government bonds, the dollar, gold, oil — all of them twitch within the same second, in a coordinated way that no single company's earnings report ever produces. That coordination is the subject of this paper. When *everything* moves at once, the cause is almost never in any one instrument. It is *macro* — short for *macroeconomics*, the study of the economy as a whole rather than of individual firms or households.

The thesis is simple and is worth stating before any detail arrives. *A small number of economy-wide variables — how fast the economy is growing, how quickly prices are rising, what it costs to borrow money, and how freely money is flowing through the financial system — set the backdrop against which every asset is priced.* Understand those four and you understand what is moving the tide, even when you cannot predict the next wave. That last clause is not a hedge; it is half the lesson. Macro is a superb lens for *context* — for knowing what kind of environment you are trading in — and a famously poor tool for *timing* the next hour or the next week. A paper that taught you the first without insisting on the second would be doing you harm.

This is also, transparently, a learning document. Macroeconomics is the author's weakest area, and the greatest risk in writing about it is not vagueness but confident error — passing a half-understood mechanism to a reader as if it were settled. The whole paper is therefore built with an unusual amount of self-imposed friction: every causal claim is stated together with how strongly it is supported, and every "this causes that" sentence is followed, wherever possible, by an example of a time it did not hold. Written this way, the paper stays useful even where it is uncertain, because the uncertainty is on the page rather than hidden.

1.2 Scope, prerequisites, and intended reader

This paper sits in the Trading Library's *Macro & Sentiment* track. It assumes *no* prior economics and *no* mathematics beyond arithmetic: everything is explained with plain numbers and simple ratios, never with calculus or statistics. Every technical term is set in *italics* and defined at its first appearance, and all of them are collected again in the glossary (Section 11). If you have never knowingly read an economic statistic in your life, you are the intended reader.

What this paper does *not* do is equally important. It does not build a model you can trade, it does not forecast the economy, and it does not tell you where rates or stocks are going. It is deliberately the most extensive document in the library precisely because it is foundational: the companion macro papers (the macro-regimes paper, on computable regimes and the central-bank

reaction function; the sentiment-and-positioning paper, on sentiment and positioning) assume everything here. After reading it, you should be able to:

- name the four master variables, say in one sentence what each measures, and read the single chart that shows it;
- explain what a *recession* is operationally and who decides one has happened;
- read an inflation number and know why the central bank might or might not care;
- look at a *yield curve* and describe its level and slope, and state honestly what an *inversion* has and has not foretold;
- explain *quantitative easing* and *quantitative tightening* in plain mechanics, and why "liquidity" is invoked to explain risk-asset moves — while knowing that link is looser than it is usually sold;
- read a week's economic calendar, know which releases tend to move markets and by roughly how much, and build a simple event map;
- and trace, step by step, how a single inflation print becomes an index move — and where that macro shock meets the order-flow world of the flow-landscape paper.

1.3 Data and reproducibility

This is a learning document, written with an unusual amount of self-imposed friction, because in a subject one is still learning the real risk is not vagueness but confident error.

One discipline runs through the paper: every causal sentence is paired with a counterexample or an explicit period in which it failed. Macro is full of relationships that are true on average and false often enough to ruin anyone who trades them as laws. Rate cuts usually support stocks — except when they are cutting *into* a collapsing economy. An inverted yield curve usually precedes a recession — except when it does not, or does so two years late. Writing the exception next to the rule is the single most important discipline in the paper.

All figures use freely available data: prices from the public Yahoo Finance endpoint, and macroeconomic series from the primary statistical agencies (the U.S. Bureau of Labor Statistics for consumer prices, the Federal Reserve's own data releases for interest rates and its balance sheet, the U.S. Treasury and the Federal Reserve Bank of New York for market and policy rates, and the National Bureau of Economic Research for recession dates). Each figure caption names its sources and its one pre-committed rule, and each is reproducible from the cited script. Where a series was, at build time, unreachable from its usual redistributor, it was pulled instead from the primary agency that publishes it; the values are identical because it is the same published data. *This is a research and educational document, not investment advice.*

1.4 Organization of this paper

The paper is complete; nothing below is a placeholder. It proceeds from the four variables, to each variable in depth, to how they combine and transmit, and finally to the limits of the whole enterprise.

- **Section 2 The four master variables** defines growth, inflation, interest rates and liquidity from zero, shows all four in one chart, and gives the simplest honest model of how they interlock — and where that model breaks.
- **Section 3 Growth and the business cycle** covers *GDP*, the phases of the cycle, the difference between *leading* and *lagging* indicators, how a recession is actually dated, and which assets tend to lead or lag — with the caveat that the timing is the hard part.
- **Section 4 Inflation from first principles** explains what inflation is, how *CPI* and *PCE* measure it and why the Fed prefers one, why it matters for the price of every asset, and the three classic stories for where it comes from — walked through the 2021–23 episode.
- **Section 5 Interest rates and the central bank** builds the *policy rate*, the *transmission mechanism*, and why rates are "the price of everything," then reads the yield curve as a picture and lays out its honest recession track record.
- **Section 6 Central banks and liquidity** covers the Fed's and ECB's mandates, *quantitative easing* and *tightening* in plain mechanics, and the much-abused idea that "liquidity" drives risk assets — with the honest counterpoint.
- **Section 7 The key data releases** treats *NFP*, *CPI*, *FOMC*, *PCE*, the *PMIs* and retail sales: what each measures, why markets react, and how large the reaction typically is, measured on free data.
- **Section 8 Cross-asset transmission** draws the standard map from rates to equities, credit, FX and commodities, and is honest about how loose the couplings are; it introduces *risk-on / risk-off* as a useful-but-crude lens.
- **Section 9 How macro shows up intraday** traces a single CPI print to an index move and connects the macro shock to the order-flow mechanics of the flow-landscape paper.
- **Section 10 What macro cannot do** is the humility section: great for context, poor for timing.
- **Section 11** collects the extensive glossary and reading ladder, followed by the references and changelog.

2 The four master variables

Almost everything that moves the whole market at once can be traced to four economy-wide quantities. They are not the only things that matter, but they are the ones large enough to move every asset together, and everything else in this paper is, in one way or another, a consequence of them. This section defines each from nothing, shows all four in a single picture, and then gives the simplest mental model of how they fit together — followed immediately by the ways that model misleads.

2.1 Growth: is the economy getting bigger?

Growth means the increase, over time, in the total quantity of goods and services an economy produces. The standard yardstick is *Gross Domestic Product (GDP)*: the market value of everything produced within a country in a given period. When people say "the economy grew 2 percent," they mean *real* GDP — GDP after stripping out the effect of rising prices — was about 2 percent larger than a year (or a quarter) earlier. The word *real* here always means "inflation-adjusted"; its opposite, *nominal*, means "in current dollars, price changes included." Growth is reported as a percentage change, and for U.S. quarterly GDP it is conventionally *annualized*: the quarter's change is expressed as the rate it would imply if it continued for a full year, which makes small quarterly numbers look larger and, in extreme quarters, enormous.

Growth is the variable that most directly justifies owning risky assets. A company's shares are, at bottom, a claim on its future profits, and profits in aggregate rise and fall with the economy. Faster growth generally means fatter corporate earnings, which is why equities as a class tend to do well in expansions. But the link from growth to stock *prices* is looser than it sounds, and the exceptions are instructive. In the second quarter of 2020 the U.S. economy suffered its sharpest contraction on record while the stock market staged one of its sharpest rallies — because prices look forward to the recovery and to the flood of policy support, not at the current quarter. And strong growth can *hurt* stocks when it arrives in an inflationary regime, because it pushes the central bank to raise rates: through much of 2022–23, good economic news was routinely bad market news. So: growth supports risk assets *on average and over time*, and the average conceals violent exceptions.

2.2 Inflation: are prices rising, and how fast?

Inflation is the rate at which the general level of prices rises over time — not the price of any one thing, but the broad tendency of prices, and therefore the rate at which the purchasing power of money erodes. If inflation is 3 percent, a basket of goods that cost 100 last year costs about 103 now, and the same money buys about 3 percent less. It is almost always quoted *year-over-year*:

this month's price level compared with the same month twelve months ago, which nets out the regular seasonal pattern of prices.

Inflation matters to markets for a reason that runs underneath everything else in this paper: it sets what the central bank does, and the central bank sets interest rates, and interest rates price every asset. High and rising inflation forces rates up; low and stable inflation lets them stay low. The single defining macro event of the 2020s — treated at length in Section 4 — was inflation rising to about 9 percent (on the seasonally adjusted measure) in mid-2022, far above the roughly 2 percent that central banks target, and the aggressive rate increases that followed. The counterexample to "inflation is always bad for stocks" is the late 1990s, when moderate inflation coexisted with a soaring market: it is not the level of inflation alone that matters but whether it is forcing the central bank's hand.

2.3 Interest rates: what does money cost?

An *interest rate* is the price of borrowing money, expressed as a percentage per year. There is not one rate but a whole structure of them, from the overnight rate the central bank controls to the yield on a thirty-year government bond. The one the news means by "the Fed raised rates" is the *policy rate* — in the United States, the *federal funds rate*, the rate at which banks lend to each other overnight, which the Federal Reserve steers to a target range. Everything else in the rate structure is built on top of it and on expectations of where it will go.

Rates are, in a real sense, the price of everything, because a dollar promised in the future is worth less than a dollar today, and the interest rate is exactly how much less — a mechanism called *discounting*, explained gently in Section 5. When rates rise, every future payment — a company's future profits, a bond's future coupons — is worth less in today's money, so asset prices tend to fall; when rates fall, the reverse. This is the closest thing macro has to a universal law, and it still has loud exceptions: in 2004–06 the Fed raised rates steadily while equities rose, because earnings were growing fast enough to more than offset the discounting drag. The direction of the rate is only half the story; what the rate is doing *relative to growth and earnings* is the other half.

2.4 Liquidity: how freely is money flowing?

Liquidity is the loosest of the four terms and the most abused, so it is worth pinning down. In the macro sense used here it means the quantity of money and near-money sloshing through the financial system, and the ease with which it can be moved into assets. The clearest single gauge is the size of the central bank's *balance sheet* — the total assets it holds — because when a central bank buys bonds it creates new money to do so, and that money ends up as *reserves* in the banking system, raising system-wide liquidity. The balance sheet ballooned during the crises of

2008 and 2020 as the Fed bought bonds on a vast scale, and it has been shrinking since 2022 as the Fed let those holdings run off.

The practitioner claim is that liquidity drives risk assets: more of it lifts stocks and credit, less of it weighs on them. There is something to this, and Section 6 gives it a fair hearing — but the honest counterpoint belongs here at the outset. The relationship is far looser than the popular "liquidity is everything" framing suggests. Japan's central bank has run an enormous balance sheet for decades without the runaway equity performance the simple story would predict, and periods of balance-sheet expansion and equity returns line up only roughly and with many exceptions. Liquidity is a real force and a poor formula.

2.5 One picture of all four

Figure 1 shows the four master variables for the United States since 1970, each on its own scale, with recessions — the grey bands, dated by the *National Bureau of Economic Research* (NBER), the semi-official arbiter introduced in Section 3 — shaded across all four. Nothing in the figure is modelled or tuned; it simply displays the four series so their scale, their volatility, and their timing relative to recessions can be seen at once.

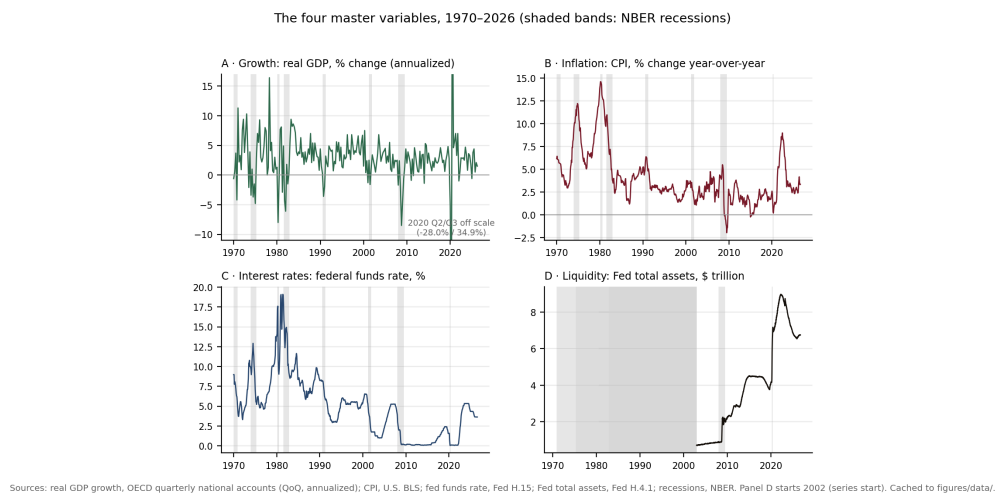


Figure 1. The four master variables of the U.S. economy, 1970–2026, each on its own scale with NBER recessions shaded in grey. (A) real GDP growth, percent at an annualized rate; (B) CPI inflation, year-over-year; (C) the effective federal funds rate; (D) the Federal Reserve's total assets in trillions of dollars, a series that begins in 2002. The 2020 output collapse and rebound are off-scale in panel A and annotated there. Reproduce with `figures/fig_e1_master_vars.py`.

Read the panels together and the texture of the macro backdrop appears. Growth (panel A) oscillates around a positive average and plunges hardest around recessions; the 2020 quarter is so extreme — a contraction of about 28.0 percent annualized, followed by a rebound of about 34.9 percent — that it is drawn off the panel's scale and labelled. Inflation (panel B) had two great

episodes in this window: the double-digit inflation of the 1970s and early 1980s, peaking near 14.6 percent in early 1980, and the post-2021 surge. The federal funds rate (panel C) tracks inflation with a lag — as high as 19 percent when Paul Volcker's Fed broke 1970s inflation, near zero after 2008 and again in 2020, then rising to about 5.3 percent in the recent tightening. And the balance sheet (panel D), which the data only cover from 2002, grew from about 0.7 trillion dollars to a peak near 9.0 trillion in 2022 and has been shrinking since — the two great liquidity expansions of 2008 and 2020 and the reduction that followed. The grey bands make the cycle visible: there were eight recessions in this span, and each one bends all four series.

2.6 The simplest honest model — and where it breaks

Here is the mental model in one paragraph, the version to carry around and to distrust in equal measure. *The economy grows; growth eventually generates inflation; the central bank raises interest rates to cool inflation; higher rates slow growth and can tip the economy into recession; the central bank then cuts rates and adds liquidity; and the cycle begins again.* This loop — growth → inflation → rates → slowdown → easing → growth — is the skeleton on which most macro commentary hangs, and it is genuinely useful as a first approximation.

It is also wrong often enough that treating it as a clock will cost you money. The links are loose and the lags are long and variable. Inflation does not follow growth reliably: the strong growth of the late 1990s came with tame inflation, while the 1970s produced *stagflation* — high inflation *and* stagnant growth together, which the tidy loop says should not happen. Rate hikes do not reliably produce recessions on schedule: the Fed raised rates through 2022–23 and, by the end of the data here, no recession had followed, the so-called soft landing. And the central bank does not always lead — sometimes it is chasing events it did not foresee, as in 2021 when it called the coming inflation "transitory" and was later forced into the fastest tightening in decades. The loop is a map of the *tendencies*; the territory is full of exceptions, and the rest of this paper is largely a tour of them.

2.7 A first look at regimes

One useful way to organize the four variables — and a preview of the macro-regimes paper — is to notice that two of them, growth and inflation, do most of the work in setting the mood of markets, and that each can be either rising or falling. Crossing those two gives four *regimes*, a crude but surprisingly durable map. When growth is rising and inflation is falling — sometimes called *goldilocks* — risk assets tend to do best: the economy is healthy and the central bank is relaxed. When growth and inflation are both rising (*reflation*), cyclical assets and commodities often lead while bonds struggle. When inflation is rising but growth is falling (*stagflation*), almost nothing works well, and this is the regime central banks most dread because their two mandated goals pull in opposite directions at once. And when growth and inflation are both falling (a

disinflationary slowdown, or in the extreme *deflation*), government bonds and cash tend to shelter capital while risk assets suffer. This quadrant is not a trading system — the boundaries are fuzzy, the regime is usually only obvious in hindsight, and the 2020s slid between all four in a few short years — but as a first mental filing cabinet it is hard to beat, and the macro-regimes paper turns it into something computable from public data with honest confidence intervals attached.

3 Growth and the business cycle

Growth is not a smooth upward line. Economies expand for a while, then stumble, contract, and recover, in an irregular rhythm called the *business cycle*. Understanding that rhythm — its phases, how it is measured, and how markets relate to it — is the first concrete skill of macro literacy.

3.1 GDP and the phases of the cycle

Recall that *GDP* is the total market value of goods and services an economy produces, and *real GDP* is that figure adjusted for inflation so that it measures quantity, not price. The business cycle is the wave-like path of real GDP (and of employment, production and income alongside it) around its long-run upward trend. It has, by convention, four phases. In *expansion*, output and employment grow; this is the normal state, and expansions typically last years. At the *peak*, growth tops out. In *contraction* — the descent — output falls and unemployment rises; if the contraction is broad and deep enough it earns the name *recession*. At the *trough*, the decline ends and the next expansion begins. The cycle is *not* periodic: expansions have ranged from about a year to more than a decade, and no one has reliably predicted their length.

3.2 What a recession actually is

Casual usage says a recession is "two consecutive quarters of falling GDP," and that rule of thumb is fine for a first pass. But it is not the official definition, and knowing the real one matters. In the United States, recessions are dated by the *Business Cycle Dating Committee* of the NBER, a committee of academic economists, which defines a recession as "a significant decline in economic activity that is spread across the economy and lasts more than a few months." They look at several monthly series — employment, real income, industrial production, spending — not GDP alone, and they date the *peak* (the last month of expansion) and the *trough* (the last month of contraction). Crucially, they announce these dates with a long lag, often a year or more after the fact, because they wait for revised data and want to avoid false calls. The 2020 recession, the shortest on record, ran from a February 2020 peak to an April 2020 trough — two months — which the "two quarters" rule of thumb would have missed entirely. The practical lesson: by the time a recession is *officially* dated, it is old news, so markets never wait for the announcement.

3.3 Leading, coincident, and lagging indicators

Because the official verdict comes so late, market participants watch a zoo of *indicators* — statistics that move with the cycle — and sort them by *when* they move relative to it. A *coincident* indicator turns at roughly the same time as the economy: *industrial production* (the physical output of factories, mines and utilities) is a classic example, and it is one of the series the NBER itself weighs. A *lagging* indicator turns after the economy has already changed direction; the unemployment rate is the textbook case, because firms are slow to hire and slow to fire. And a *leading* indicator tends to turn *before* the economy — these are the ones forecasters covet. The slope of the yield curve (Section 5) is the most famous leading indicator; others include building permits, new orders, and the stock market itself, which is often described, only half in jest, as having "predicted nine of the last five recessions." That joke is the whole difficulty in one line: leading indicators lead *on average*, with a lead time so variable and a false-positive rate so high that acting on them is genuinely hard.

The labor market deserves a special word, because it is the read on growth that markets and central banks watch most closely, and because it hides a subtle trap. Employment — how many people have jobs — is where a slowing economy shows up in the most human terms, and the monthly U.S. *employment report* (Section 7) is often the single largest scheduled market event of the month. Confusingly, its two headline numbers sit on *different* sides of the cycle. The count of jobs added each month is roughly *coincident*, moving with the economy in near-real time. The *unemployment rate*, by contrast, is a *lagging* indicator: firms are slow to lay workers off when activity first softens and slow to rehire when it recovers, so unemployment often keeps rising for months after a recession has technically ended. This is why a low, falling unemployment rate is not the all-clear it appears to be — historically it has been at its lowest *just before* recessions, not after them, precisely because a fully-employed economy is one running at the top of its cycle. The recurring lesson holds even here: the most-watched gauge of the economy's health tells you most clearly about where it has already been.

Figure 2 shows the two kinds side by side for the United States. The top panel is industrial production, year-over-year — a coincident gauge that visibly turns down into every recession. The bottom panel is the *10-year minus 3-month Treasury spread*, a leading gauge: the difference between the interest rate on a 10-year government bond and a 3-month one, which tends to fall below zero (an *inversion*, Section 5) before recessions.

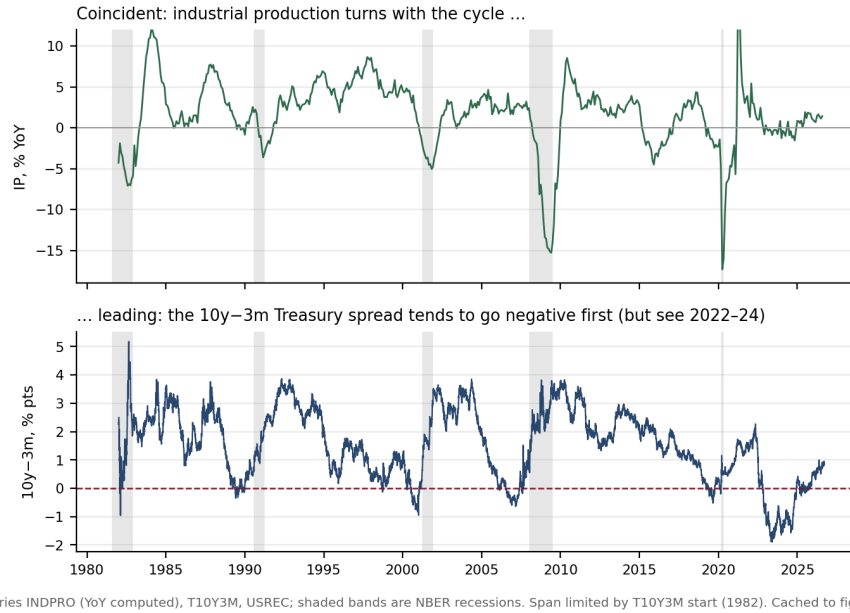


Figure 2. The business cycle as data, 1982–2026. Top: industrial production, year-over-year percent change — a coincident indicator that turns down with each recession (grey bands, NBER-dated). Bottom: the 10-year minus 3-month Treasury yield spread in percentage points — a leading indicator that has usually fallen below zero (the dashed line) before a recession, with the large and still-unresolved inversion of 2022–24 visible at the right. Reproduce with `figures/fig_e1_cycle.py`.

The figure is honest about the messiness. For each recession start in the window the script checks whether the 10-year-minus-3-month spread had gone negative in the prior eighteen months, and records how weak industrial production was at the start. The spread had gone negative before four of the five recessions in this window — those beginning in 1990, 2001, 2008 and 2020 — the lone exception being the 1981–82 recession, which begins just as the 3-month constant-maturity series itself starts and so cannot be assessed. Industrial production, the coincident gauge, gave a mixed reading at those starts: it was already contracting in some (about –2 percent year-over-year in 2001, about –5 percent in 2020) and still positive in others (1990 and 2008), so even the "turns with the cycle" indicator did not turn cleanly on cue. And, once again, the most recent inversion — with the spread last below zero in late 2025 — had fired without a dated recession following by the end of the data. The point is not that the leading indicator failed — it usually fired — but that "usually," "before," and "by how long" are doing an enormous amount of work, and none of them is a schedule you can trade on.

3.4 Which assets lead and lag the cycle

There is a standard, textbook picture of how asset classes rotate through the cycle, and it is worth knowing precisely because it is so often oversold. In the stylized story: the stock market and other risk assets tend to *lead*, bottoming before the recession ends and peaking before it begins, because prices discount the future; long-term *government bonds* tend to do well as growth and rates fall

into a slowdown; *commodities*, tied to current industrial demand, tend to lag, peaking late in an expansion; and defensive sectors outperform going into weakness while cyclical sectors outperform coming out. This "rotation" framework is a useful vocabulary. It is also far too clean: the lead times vary enormously, the ordering breaks in any given cycle, and — the essential caveat — knowing that stocks *tend to* bottom before the economy does not tell you *which* month. The 2020 low came weeks after the recession began; the 2007 market peak came two months before the recession's official start; there is no reliable clock. Cyclical rotation is a lens for understanding *what has happened*, not a timing tool for *what will*.

4 Inflation from first principles

Of the four master variables, inflation is the one that has most directly driven markets in the 2020s, and it is the one most surrounded by confusion. This section builds it carefully.

4.1 What inflation is, and how it is measured

Inflation is the rate of increase of the general price level. To measure it you need a *price index*: a single number that tracks the cost of a fixed, representative *basket* of goods and services over time. The best-known is the *Consumer Price Index (CPI)*, published monthly by the Bureau of Labor Statistics, which tracks what a typical urban household pays for food, housing, transport, medical care, and so on. The index itself is just a level — set to 100 in a base period — and inflation is its rate of change, almost always quoted *year-over-year*: the index this month divided by the index twelve months ago, minus one, as a percentage.

Two distinctions matter immediately. First, *headline* versus *core*. Headline inflation includes everything; *core* inflation strips out food and energy prices, which are volatile and swing on weather, harvests and geopolitics. Core is not "the inflation that matters to your grocery bill" — it obviously understates that — but a cleaner read on the *underlying trend*, which is what a central bank steering a slow ship cares about. Second, *seasonal adjustment*. Prices have a regular within-year pattern (gasoline in summer, gifts in December), and *seasonally adjusted* series remove that predictable wiggle so that a genuine change is not confused with the calendar. The widely-quoted "9.1 percent" peak of 2022 was the *not*-seasonally-adjusted headline figure; the seasonally adjusted series, which the figures here use, peaked near 9.0 percent. Neither is wrong; they answer slightly different questions, and quoting one as if it were the other is a common, avoidable error.

4.2 CPI versus PCE, and why the Fed prefers PCE

There is a second major inflation gauge, and the difference between the two is a small but genuine piece of literacy. Alongside CPI sits the *Personal Consumption Expenditures* price index

(*PCE*), produced by the Bureau of Economic Analysis. It measures much the same thing — the rising cost of what people consume — but with a broader scope and, importantly, a formula that lets the basket *shift* as people substitute away from goods whose prices rise. Because of that substitution and other methodological differences, *PCE* inflation typically runs a few tenths of a percentage point *below* *CPI*. The Federal Reserve states its 2 percent inflation goal in terms of the *headline PCE* price index, not *CPI*, for exactly these reasons: it considers *PCE*'s broader, substitution-aware basket a better measure of the cost of living people actually experience. It watches *core PCE* — *PCE* excluding food and energy — as a less-noisy guide to the underlying trend, but the 2 percent target itself is written on headline *PCE*. For a market participant the practical consequences are two: the Fed's "2 percent target" is a *PCE* 2 percent, which is a slightly higher number in *CPI* terms; and *CPI*, released earlier each month, is the one that moves markets on the day, while *PCE*, released later, is the one the Fed formally weighs. Section 7 returns to both as scheduled events.

4.3 Why inflation matters for the price of every asset

Inflation reaches markets through two doors. The first is *purchasing power*: money loses value at the rate of inflation, so an asset that merely holds its nominal value is losing real ground, and investors demand higher returns to compensate — which means higher interest rates. The second, larger door is the central bank: its job (Section 5, Section 6) is to keep inflation low and stable, so when inflation rises it raises rates, and when inflation falls it can cut. Since interest rates price every future cash flow through *discounting* (Section 5.2), inflation ends up embedded in the price of every stock, bond, and building. This is why an inflation surprise moves the *whole* market at once and in a coordinated way: it is really a surprise about the future path of interest rates. The direction is not mechanical, though — the counterexample is any period where inflation rose but the central bank was expected to *tolerate* it, in which case bonds sell off but the rate-hike fear that hurts equities never fully materializes. What matters is not inflation in isolation but inflation *as read through the central bank's likely reaction*.

4.4 Where inflation comes from: three stories

Economists tell three overlapping stories about the *source* of inflation, and each carries a different implication.

- *Demand-pull* inflation is "too much money chasing too few goods": demand outruns the economy's capacity to produce, and prices are bid up. A booming economy near full employment generates it.
- *Cost-push* inflation comes from the supply side: a jump in the cost of a key input — oil, shipping, wages — pushes prices up even without excess demand. The 1970s oil shocks are the archetype.

- *Monetary* inflation is the longer-run idea, associated with Milton Friedman's dictum that "inflation is always and everywhere a monetary phenomenon": sustained inflation ultimately requires the money supply to grow faster than output.

These are not rival theories so much as different contributors that can operate at once, and disentangling them in real time is genuinely hard — which is exactly why central banks and markets so often disagree about whether a given burst of inflation will fade or persist. The honest caveat is that the profession's forecasting record here is poor: the 2021 inflation was widely, and wrongly, expected to be transitory cost-push that would fade on its own, and instead it broadened into something more persistent. Getting the *story* wrong is not an academic slip; it drove one of the largest central-bank policy errors in decades.

4.5 The 2021–23 episode as the modern teaching case

The recent inflation is the best classroom the subject has offered in forty years, because every element above appears in it. Figure 3 tells the story in one chart: headline and core CPI inflation on the left axis, and the central bank's policy-rate response on the right.

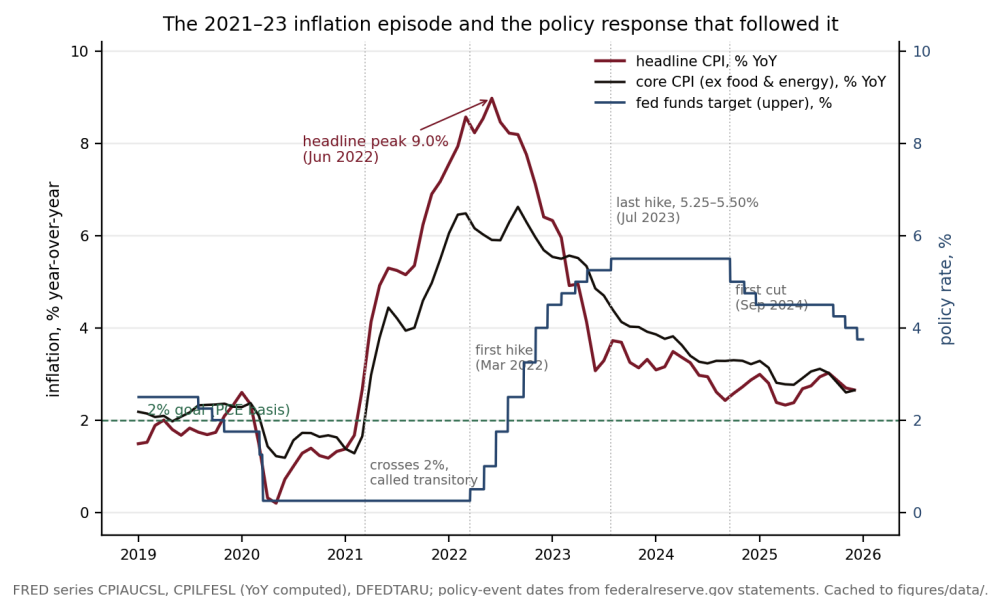


Figure 3. The 2021–23 inflation episode and the policy response. Headline CPI inflation and core CPI inflation (both year-over-year, left axis) against the upper limit of the federal funds target range (right axis, in percent). Headline inflation first crossed 2 percent in 2021, was initially called transitory, peaked near 9 percent (seasonally adjusted) in mid-2022, and then fell; the policy rate was held near zero into 2022, then raised rapidly to a peak of 5.50 percent by mid-2023, and cut from late 2024. Reproduce with `figures/fig_e1_inflation_episode.py`.

The sequence, with the dates the script computes from the data, is a compressed tour of the whole subject. Headline inflation first pushed above the 2 percent goal in March 2021, driven at first by

pandemic supply snarls and a demand rebound — a textbook mix of cost-push and demand-pull. It was widely called *transitory*. It was not: headline CPI climbed to about 9.0 percent (seasonally adjusted) by June 2022, with core peaking near 6.6 percent a few months later, in September 2022. The policy response, initially delayed, then became the most aggressive in decades — the target range's upper limit went from near zero to a peak of 5.50 percent by July 2023, roughly thirteen months after inflation first cleared the target. Inflation then fell substantially without (through the end of the data here) a recession, and the first rate cut came in September 2024. Whether this counts as a triumphant *soft landing* or a lucky escape is still debated, and the paper takes no side; what it illustrates cleanly is the whole loop of Section 2.6 playing out once, at high speed, with a visible forecasting error at its heart.

4.6 How to read a CPI report

Because CPI is the release that most often moves the whole market (Section 7), a little structural literacy pays off, and it costs only a paragraph. A CPI report is not one number but a stack of them. The largest single component is *shelter* — the cost of housing — which is both large and slow-moving, so it dominates the trend and lags turning points by many months. Beneath the headline, analysts split the basket into *goods* and *services*, and services further into *core services*, because these behave very differently: goods prices swung wildly with supply chains in 2021–22 and then fell outright, while services inflation, tied to wages, proved far stickier and came down only slowly. Reading *which part* of the basket is driving a given month tells you far more about whether inflation will persist than the headline figure alone does.

Two traps catch newcomers, and both are worth naming. The first is the *base effect*: a year-over-year inflation rate can fall simply because the reading *twelve months earlier* was unusually high, even while prices are still rising briskly month to month — the movement is in the comparison base, not in the present. Watching the month-over-month change (often annualized) guards against being fooled by it. The second is confusing *disinflation* with *deflation*. Disinflation — the whole story of 2023–24 — means inflation is *slowing* while prices still *rise*; deflation means prices actually *fall*, a rarer and in some ways more dangerous condition, because falling prices can feed on themselves as buyers wait for lower ones, as Japan's long experience showed. A market cheering that "inflation is coming down" is almost always cheering disinflation, not deflation, and the two have opposite implications for policy.

5 Interest rates and the central bank

Interest rates are the hinge of the whole system: the variable the central bank actually controls, and the one through which growth and inflation reach every asset price. This section builds the policy rate, explains discounting in the gentlest possible terms, and then reads the yield curve.

5.1 The policy rate and how it is set

A *central bank* is a public institution charged with managing a currency and the monetary system. The United States' is the *Federal Reserve* ("the Fed"); the euro area's is the *European Central Bank* (ECB). The lever each pulls is a very short-term interest rate. In the U.S. it is the *federal funds rate* — the rate at which banks lend reserves to one another overnight. The Fed does not decree it by fiat; it sets a *target range* (a band, e.g. 5.25 to 5.50 percent) and uses its tools to keep the actual traded rate, the *effective federal funds rate*, inside that band. The committee that decides is the *Federal Open Market Committee* (FOMC), which meets eight times a year on a published calendar (Section 7). When the news says "the Fed hiked by 25 basis points," it means the FOMC raised the target range by a quarter of a percentage point — a *basis point* being one hundredth of a percent, the unit rates are quoted in.

Why does an overnight rate between banks matter to anyone else? Because it anchors the whole structure of rates. Every other interest rate — what a business pays to borrow, what a mortgage costs, the yield on a government bond — is built on top of the expected path of the policy rate plus various premiums. Move the anchor, and, with varying force and lag, the whole chain moves.

5.2 Discounting: why rates are the price of everything

Here is the one idea that makes rates central, stated with arithmetic only. Money in the future is worth less than money now, because money now can be invested to earn interest. If the interest rate is 5 percent, then 105 dollars a year from now is worth exactly 100 dollars today — you would be indifferent between them, since 100 today grows to 105. Turn that around: to find what a future dollar is worth today, you *discount* it by the interest rate. At a 5 percent rate, one dollar a year out is worth about 95 cents today; two years out, about 91 cents; and so on, shrinking further the higher the rate.

Now notice what an asset *is*. A stock is a claim on a company's future profits; a bond is a claim on future coupon and principal payments; a rental property is a claim on future rents. Every asset is a bundle of future dollars, and its fair price today is the sum of those dollars each discounted back to the present. So when the interest rate rises, the discount is heavier, every future dollar is worth less today, and the fair price of *every* asset falls — mechanically, before any change in the profits or rents themselves. This is the deepest reason a change in rates moves the whole market at once, and it is why "rates are the price of everything" is more than a slogan. The honest qualification, from Section 2.3, still applies: rising rates lower prices *all else equal*, and all else is frequently not equal — if profits are growing fast enough, as in the mid-2000s, prices can rise even as rates do.

5.3 The transmission mechanism

The chain from the policy rate to the real economy is called the *transmission mechanism*, and naming its links demystifies a lot of commentary. A change in the policy rate feeds, with lags of months to years, into: *borrowing costs* (loans, mortgages, corporate bonds become dearer or cheaper), which changes how much households and firms spend and invest; *asset prices* (via the discounting of Section 5.2), which changes wealth and confidence; the *exchange rate* (higher rates tend to attract capital and strengthen the currency, Section 8), which changes the price of imports and exports; and *expectations* themselves, since what people believe the central bank will do shapes their behavior today. All of this eventually bears on growth and inflation, closing the loop back to Section 2.6. The essential and repeatedly-learned caveat is Milton Friedman's: monetary policy acts with "long and variable lags," so the effect of a rate change shows up unpredictably far in the future — which is precisely why central banks are so often accused of moving too late or too far.

5.4 The yield curve as a picture

The *yield* of a bond is the annual return it offers if held to maturity; a *government bond* (a *Treasury* in the U.S.) is a loan to the government, considered free of default risk, so its yield is close to a pure interest rate for its maturity. Plot the yield of government bonds against their *maturity* — 3 months, 2 years, 10 years, 30 years — and you get the *yield curve*, a single picture that encodes what the market thinks about rates and growth. Two features carry most of the meaning. Its *level* is how high rates are in general, dominated by the current and expected policy rate. Its *slope* is whether long-term yields are above or below short-term ones. Normally the curve *slopes upward*: lending for longer earns more, both to compensate for tying money up and because the economy is expected to grow.

The signal traders watch is when the slope goes the *wrong* way. An *inversion* — long-term yields *below* short-term ones — means the market expects the central bank to be *cutting* rates in the future, which is usually a bet that the economy will weaken enough to force cuts. That is why an inverted curve has a reputation as a recession warning. The academic case for it is real: Estrella and Mishkin (1998) found the yield-curve slope to be among the best single financial predictors of U.S. recessions at horizons beyond a quarter.

5.5 The honest track record of the inversion signal

Reputation is one thing; the record is another, and this is exactly the kind of claim the paper is obliged to test rather than repeat. Figure 4 plots the *10-year minus 2-year* Treasury slope since 1976 — the most-watched inversion measure — with every episode below zero shaded and recessions marked, and the accompanying script prints, for each inversion, how long until the next recession began.

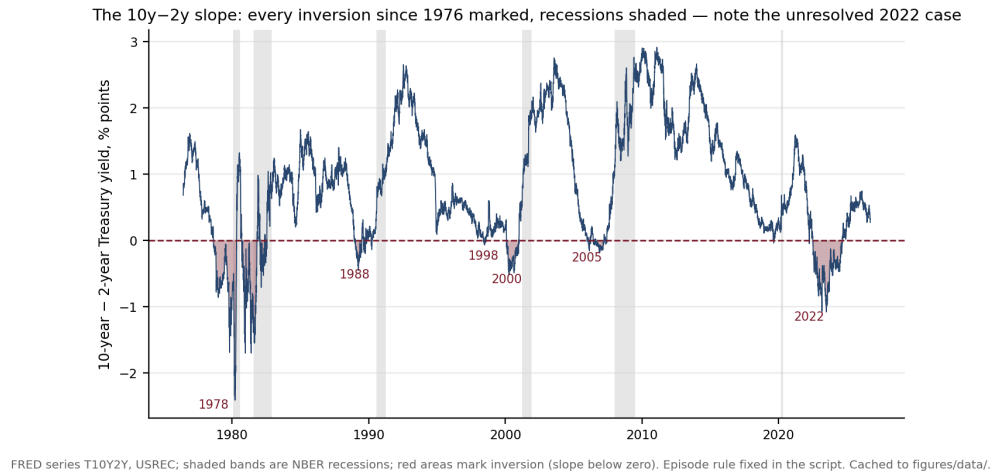


Figure 4. The 10-year minus 2-year Treasury yield slope, 1976–2026, in percentage points, with inversions (slope below zero) shaded red and NBER recessions shaded grey. Every recession in the span was preceded by an inversion, but the lead time from the start of the inversion to the recession ranged from about one year to nearly three, and the large inversion that began in 2022 had not been followed by a recession by the end of the data. Reproduce with `figures/fig_e1_yield_curve.py`.

The numbers, reported as they fall out of one pre-committed rule for what counts as an inversion episode, are a model of why macro signals are so treacherous. The sustained inversions that began in 1978, 1980, 1988, 2000, 2005–06 and 2019 were each followed by a recession — a genuinely impressive hit rate. But the *lag* from the inversion's start to the recession's start was 18 months in the late-1970s case, about 20 months in 1988, 14 months in 2000, roughly 25 months in 2005–06, and only 7 months for the brief 2019 signal. A "warning" whose lead time ranges from half a year to more than two years is nearly useless for timing, even when it is reliable in direction. And then the essential counterexample: the curve inverted deeply in 2022 and, at the end of the data here — more than four years and over fifty months later — no recession had been dated. The signal that "always" precedes recessions had, in the most recent and most dramatic case, either failed or been running the longest lag on record. That is the honest state of the most famous leading indicator in macro: excellent as context, dangerous as a clock.

5.6 Real versus nominal rates, and the term premium

Two refinements complete the picture of rates, and both turn up constantly in commentary. The first is the distinction between the *nominal* interest rate — the rate actually quoted, say 4 percent — and the *real* interest rate, which is roughly the nominal rate minus expected inflation. If a bond yields 4 percent and inflation is expected to run at 3 percent, the *real* return is only about 1 percent; that real rate, not the nominal one, is what actually governs saving, borrowing and investment, because it measures the true cost of money after the erosion of purchasing power. The bond market even lets you read the market's inflation expectation off a screen: the U.S. Treasury issues inflation-protected bonds (*TIPS*), whose yield is a real rate, and the gap between an

ordinary bond's nominal yield and the TIPS yield of the same maturity is the *breakeven inflation* rate — the market's implied forecast of average inflation over that horizon. When commentators say "inflation expectations remain well anchored," this breakeven is usually the number they mean.

The second refinement is the *term premium*: the extra yield investors demand for holding a long-term bond instead of rolling over short-term ones, as compensation for the risk that rates move against them over the long wait. It cannot be observed directly — it has to be estimated — and it matters because it means the yield curve's slope (Section 5.4) reflects *two* things at once: the market's expected path for the policy rate, and this premium. That ambiguity is exactly why the inversion signal is noisier than it looks. A curve can invert because the market expects rate *cuts* (a genuine recession warning) or because the term premium has simply fallen (a more technical, benign cause), and telling the two apart in real time is genuinely hard. Some economists argued precisely this about the deep 2022 inversion — that an unusually compressed term premium made it a weaker recession signal than history implied — and, since no recession had been dated by the end of the data here, that more cautious reading has not obviously been proven wrong.

6 Central banks and liquidity

Rates are the central bank's everyday tool. In crises it reaches for a second, blunter one — changing the *quantity* of money directly — and this is where the much-invoked idea of "liquidity" comes from. This section covers what central banks are trying to do and how the balance sheet works, with the honesty the topic badly needs.

6.1 Mandates: what central banks are for

A central bank acts under a *mandate*, a legal statement of its goals. The Federal Reserve has a *dual mandate*: maximum employment *and* stable prices (operationalized as 2 percent inflation on the headline PCE price index). The European Central Bank has, by treaty, a *primary* objective of price stability, with support for the wider economy explicitly subordinate to it. This difference is not trivia: it shapes how each reacts to the same shock. Facing high inflation *and* rising unemployment — stagflation — the Fed must weigh two goals it cannot serve at once, while the ECB's hierarchy tells it to prioritize inflation. Knowing an institution's mandate is the first step to anticipating its *reaction function*, the systematic way it responds to data, which is the whole subject of the macro-regimes paper.

6.2 Quantitative easing and tightening, in plain mechanics

When the policy rate is already near zero, a central bank cannot cut much further, so to add more stimulus it turns to *quantitative easing* (QE). The mechanics are simpler than the acronym

suggests. The central bank creates new money (electronically) and uses it to *buy* bonds — mostly government bonds — from banks and investors in the open market. Three things follow. The sellers now hold cash (reserves) instead of bonds, raising system *liquidity*. The buying pushes bond prices up and their yields *down*, lowering long-term interest rates that the policy rate alone cannot reach. And the central bank's *balance sheet* — the assets it holds — swells by the amount bought, which is why balance-sheet size (Figure 1, panel D) is the standard liquidity gauge. *Quantitative tightening* (QT) is the reverse: the central bank lets bonds mature without replacing them (or sells them), which drains reserves, nudges long yields up, and shrinks the balance sheet. The Fed ran huge QE programs after 2008 and in 2020, and has been running QT since 2022.

6.3 Does liquidity move risk assets? The honest version

The popular claim — sometimes compressed to "don't fight the Fed" or "liquidity is everything" — is that expanding the balance sheet lifts stocks and credit, and shrinking it weighs on them. The intuition is not empty: QE lowers the discount rate on every asset (Section 5.2), pushes investors out of low-yielding bonds into riskier assets in a "reach for yield," and signals that the central bank stands ready to support markets. Over 2009–2021, expanding balance sheets and rising equity markets did broadly coincide.

But the paper's rule is to put the counterexample right next to the claim, and here it is a strong one. The balance-sheet-to-equity link is far looser than the meme implies, and reads too much causation into a coincidence. The Bank of Japan has held an enormous balance sheet — larger relative to its economy than the Fed's — for well over a decade without producing the runaway equity returns the simple story predicts. Within the U.S. record, the correlation between balance-sheet changes and equity returns is loose and unstable over any window short enough to trade: there are stretches where the balance sheet grew and stocks fell, and stretches of the reverse. Attributing every rally to "liquidity" is a classic case of a story that explains everything after the fact and predicts little before it. Liquidity is a genuine force on risk appetite and a poor mechanical formula for returns — which is why the more careful treatment in the macro-regimes paper reaches for *financial conditions indices* (a broad composite of rates, spreads, the dollar and equity levels) as a cleaner macro-to-markets link than any single quantity, balance sheet included.

6.4 The ECB contrast, and where liquidity actually bites

Two closing observations sharpen the picture. First, the institutional contrast is not trivia. Because the ECB's mandate ranks price stability *above* support for growth (Section 6.1), and because it must operate across many national bond markets rather than one, its version of quantitative easing came later than the Fed's in the 2010s and carried extra legal and political constraints the Fed never faced. The same tool, in a different institutional setting, works differently — a standing reminder that "central banks do X" is always shorthand for *a specific* bank acting under *specific*

rules, and that transplanting an intuition from the Fed to another central bank is a common way to be wrong.

Second, it is worth being concrete about *where* liquidity actually reaches markets, because "the link is loose" (Section 6.3) is not the same as "it does nothing." The cleanest-documented channel is the *portfolio-balance* effect: when the central bank buys safe long-term bonds and takes them out of the market, the investors who sold them are left holding cash they must redeploy, and some of it flows into riskier assets, gently lifting their prices and compressing their risk premia. The effect is real but modest, and it is hard to isolate because a dozen other forces are always moving at the same time. That is the responsible version of the liquidity story: a genuine, limited mechanism — not a dial that sets the level of the stock market. It also explains why the most careful empirical work has drifted away from the raw balance-sheet number toward the broader financial-conditions composite the macro-regimes paper takes up.

7 The key data releases

Macro reaches the trading day through a calendar of scheduled *data releases* — official statistics published at pre-announced times. A handful of them reliably move the whole market, and knowing which, and roughly how much, is the most directly practical macro skill. This section catalogues the important ones and shows how to build a week's event map.

7.1 The releases that matter, and what each measures

The following are the releases that most consistently move U.S. markets.

- The *employment report* (or *non-farm payrolls*, *NFP*), released monthly by the Bureau of Labor Statistics, reports how many jobs the economy added and the unemployment rate. As the timeliest broad read on growth, it is often the single biggest scheduled mover of the month.
- The *Consumer Price Index* (*CPI*), monthly, is the first major inflation reading each month and, in an inflation-focused regime, routinely the largest mover of all — because it most directly shapes expectations for the Fed.
- The *FOMC decision*, eight times a year, is the rate decision itself, plus the statement, the projections, and the Chair's press conference. Even when the rate is unchanged, the *guidance* about the future path moves markets.
- The *PCE* price index, monthly, is the Fed's preferred inflation gauge (Section 4.2); it is released later than CPI and less often surprises, but it is the one the Fed formally targets.
- The *Purchasing Managers' Indexes* (*PMIs*), including the U.S. *ISM* surveys, are monthly diffusion indexes built from surveys of company purchasing managers; a reading above 50

means the sector is expanding, below 50 contracting. As *soft* data (survey-based) they arrive fast and are watched as leading signals.

- *Retail sales*, monthly, measure consumer spending, the largest component of the economy, and move markets when they diverge from expectations about the consumer's health.

A recurring subtlety runs through all of them: markets trade the *surprise*, not the *level*. Expectations are set in advance (by economist surveys), largely priced in, and the reaction is driven by the *gap* between the released number and the expected one. A "good" number that everyone already expected can move nothing; a "bad" number far better than feared can rally the market. This is why the reaction to a release so often seems backwards to newcomers.

7.2 How large is the reaction? A measurement

It is one thing to assert that these days are "big" and another to measure it. Figure 5 does the honest, minimal version for the most identifiable of these events — scheduled FOMC decision days — using free daily S&P 500 data and one pre-committed comparison: the average *absolute* daily move on the 120 scheduled FOMC decision days since 2011 versus all other trading days.

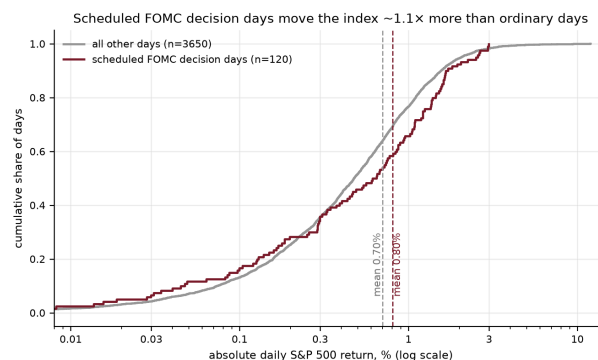


Figure 5. Distribution of absolute daily S&P 500 percentage moves on scheduled FOMC decision days (120 days, 2011–2026) versus all other trading days (about 3,650 days), shown as cumulative curves on a logarithmic scale. The average absolute move is about 0.80 percent on FOMC decision days against about 0.70 percent on ordinary days; measured at the median the gap is proportionally larger, and a move exceeding 1 percent is much more common on decision days. One pre-committed comparison, reported as-is. Reproduce with figures/fig_e1_event_days.py.

The numbers are deliberately unglamorous, and that honesty is the point. On the 120 scheduled FOMC decision days in the sample the average absolute move of the S&P 500 was about 0.80 percent, against about 0.70 percent on the roughly 3,650 other days — a ratio of about 1.14. By the *median*, the contrast is sharper: 0.63 percent versus 0.48 percent, a ratio near 1.32, and moves larger than 1 percent occurred on about 34 percent of FOMC days against 23 percent otherwise. The single largest FOMC-day move in the window was about 3.0 percent, on 4 May 2022. So decision days *are* bigger, clearly and measurably — but by a modest multiple, not a spectacular

one, and the averages are dragged around by a few outliers. This is a small, useful, defensible fact of exactly the kind macro can honestly supply: the calendar tells you *when* volatility is more likely, not *which way* it will go. (The comparison excludes the unscheduled emergency actions of March 2020, because the question asked is about the *published* calendar a trader can know in advance.)

7.3 Building a week's event map

The practical routine is simple and worth making a habit. At the start of each week, list the scheduled releases with their exact times and the consensus expectation for each, and mark the ones with a history of moving the whole market — CPI, NFP, and any FOMC day above all. That list is your *event map*: it tells you when liquidity is likely to thin out ahead of a number, when a surprise could reprice everything at once, and when the tape is likely to be quiet. It is not a set of trades; it is a map of *when the ground may shift*, and its main use is defensive — sizing down or standing aside into a major release rather than being surprised by it. This is the natural bridge to the intraday world of Section 9 and to the flow-landscape paper on order flow.

To make it concrete, a typical week's map might read: *Monday* quiet, with no first-tier release; *Tuesday* the CPI report at 8:30 a.m. Eastern — the week's main event, with the whole market likely to reprice on the surprise; *Wednesday* an FOMC decision at 2:00 p.m. followed by the Chair's press conference at 2:30, the second big event; *Thursday* the weekly jobless-claims number (a minor, high-frequency labor read) and perhaps a PMI; *Friday* quiet into the weekend. From that single sketch a trader already knows the *shape* of the week: risk is concentrated on Tuesday morning and Wednesday afternoon, liquidity will likely thin in the minutes before each, and the sensible default is to carry less exposure into those two windows and more on the quiet days. Notice what the map deliberately does *not* contain: any claim about *direction*. It marks *when* the ground may move, not *which way* — which is the honest boundary of everything the calendar can offer.

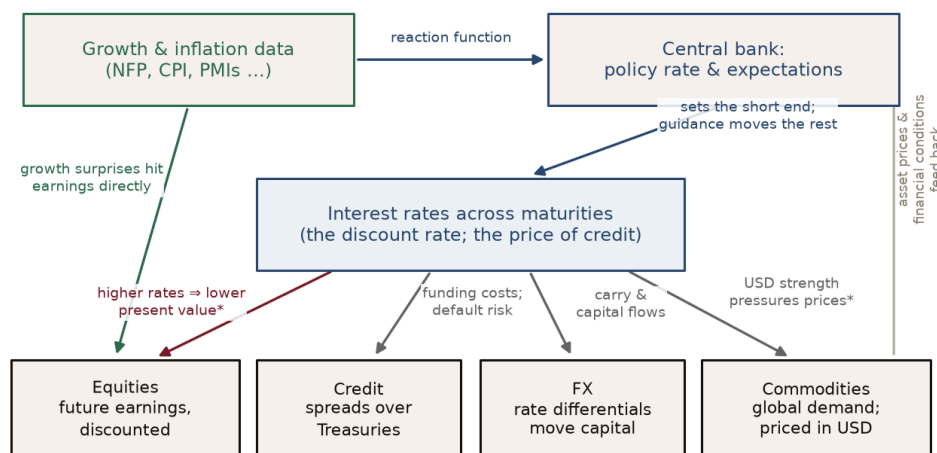
8 Cross-asset transmission

A macro shock rarely stays in one asset class. A change in rates or rate expectations ripples outward — to equities, to corporate credit, to currencies, to commodities — in a pattern loose enough to mislead and regular enough to be worth learning. This section draws the map and, as always, marks where it fails.

8.1 The standard linkages

Figure 6 is a deliberately simple schematic of the textbook transmission, and it should be trusted only as far as the banner across its foot allows.

Cross-asset transmission: the textbook map (to be trusted only loosely)



* Textbook direction, all else equal — every coupling here is loose and regime-dependent. Counterexample: in 2022 equities and bonds fell together; in 2020–21 rising rates coexisted with rising equities.

Figure 6. A textbook map of cross-asset transmission. Growth and inflation data drive the central bank's reaction function; the central bank sets the short-term rate and, through guidance, influences the whole curve of interest rates; and rates then propagate to equities (through the discounting of future earnings), to credit (through funding costs and default risk), to foreign exchange (through rate differentials that move capital), and to commodities (through global demand and the dollar). The banner records the essential caveat that every coupling shown is loose and regime-dependent. Reproduce with `figures/fig_e1_transmission.py`. (Didactic; no data download)

Trace the arrows in the textbook direction, all else equal. *Rates to equities*: higher rates discount future earnings more heavily, so equities tend to fall (Section 5.2). *Rates to credit*: higher rates raise borrowing costs and default risk, so *credit spreads* — the extra yield corporate bonds pay over government bonds — tend to widen, meaning corporate debt underperforms. *Rates to FX*: higher rates in one country tend to attract foreign capital seeking the higher return, strengthening its currency (a mechanism called *carry*). *Rates to commodities*: a stronger dollar makes dollar-priced commodities more expensive abroad and tends to weigh on their prices, while the level of global growth pulls the other way through demand. Each arrow is a real tendency with a real mechanism behind it.

8.2 How loose the couplings really are

Now the discipline. Every one of those arrows is loose, and each has famous exceptions — which is why the figure's own banner says so. The rates-to-equities link failed spectacularly in the mid-2000s and late 1990s, when rising rates coexisted with rising stocks because earnings growth dominated. The comforting idea that government bonds *hedge* equities — that when stocks fall,

bonds rise — broke down badly in 2022, when rising rates drove stocks *and* bonds down together, one of the worst years on record for the classic balanced portfolio. Currency moves routinely defy rate differentials for months when other forces (risk sentiment, trade flows, intervention) dominate. The couplings are directional tendencies that hold *often enough to be worth knowing and seldom enough to be dangerous to trade mechanically* — the recurring shape of every macro relationship in this paper.

A single worked example shows the map and its looseness at once. Suppose the central bank delivers a *hawkish* surprise, signalling that rates will stay higher for longer than the market expected. In the textbook chain, short-term yields rise first; the whole curve lifts with them; equities fall as future earnings are discounted more heavily; the currency strengthens as the higher rates draw in foreign capital; gold, which pays no interest, tends to soften because the opportunity cost of holding it has gone up; and credit spreads widen a little as tighter policy raises perceived default risk. On many days that is exactly the sequence, and being able to read it in real time is a genuine skill. But on other days the *same* hawkish surprise sends equities *up*, because the market reads the central bank's resolve as confidence in the economy's strength; or sends the currency *down*, because traders conclude the extra hikes will eventually force a recession and then rate cuts. The map gives you the base case; the prevailing regime and the way participants are already positioned decide whether reality follows it.

8.3 Risk-on and risk-off

There is a cruder but genuinely useful lens that sits on top of all this: *risk-on / risk-off*. On a *risk-on* day, investors want return: equities, corporate credit, cyclical commodities and higher-yielding currencies rise together, while safe havens — government bonds, the dollar, the yen, gold — lag. On a *risk-off* day, investors want safety: the safe havens rise and the risk assets fall together, in the coordinated way that opened this paper. Much of the day-to-day co-movement across otherwise unrelated markets is just this single dial swinging. It is a lens, not a law: the "safe havens" do not always cooperate (the dollar and Treasuries did *not* reliably rally in the 2022 sell-off), and labelling a day risk-off explains the co-movement without predicting the next one. Used as vocabulary it clarifies; used as a signal it disappoints.

9 How macro shows up intraday

The four master variables move on a horizon of months and quarters, but they reach the trading screen in seconds. This short section connects the two timescales and hands off to the order-flow papers.

Follow a single CPI release through the chain. At 8:30 a.m. Eastern the number prints; suppose it comes in three-tenths of a percent above the consensus expectation. Instantly, the *surprise*

(Section 7.1) is read as "inflation is stickier than thought," which is read as "the Fed will keep rates higher for longer," which — through the discounting of Section 5.2 — lowers the fair value of every future cash flow at once. Bond yields jump; equity-index futures drop; the dollar rises on the wider rate differential (Section 8). None of this waits for a committee or a forecast: it is thousands of participants repricing the *expected path of policy* in real time, which is why the whole board moves together in the same instant. The size of that move is, on average, the modest multiple measured in Figure 5 — bigger than an ordinary day, but not limitless.

This is exactly where macro meets the microstructure of the flow-landscape paper. The *direction* of the initial repricing is macro — set by the surprise and its implication for rates. But the *path* the price takes in the minutes that follow — the overshoot, the fade, the air-pockets and the snap-backs — is order flow: who is forced to hedge, where liquidity is thin, which options dealers must trade to stay balanced. A macro catalyst lights the fuse; market microstructure shapes the explosion. A trader who understands only the macro knows *why* the market gapped and is then lost in the noise; a trader who understands only the flow sees the gap and does not know what lit it. The two levels are complements, and the honest boundary of *this* paper is that it can explain the catalyst and the average size of the reaction, and can say almost nothing reliable about the path — which is the subject of other papers.

10 What macro cannot do

Every claim in this paper has come with a caveat; this section collects them into a single, deliberate statement of the limits, because for a self-taught author the temptation to overreach is the real danger.

Macro is excellent for *context* and poor for *timing*, and the gap between those two is where most money is lost. Consider what the previous sections actually established. The yield curve inverts before recessions — with a lead time ranging from seven months to more than two years, and at least one recent case where no recession followed at all (Section 5.5). Liquidity influences risk appetite — through a link so loose that attributing any particular rally to it is usually storytelling (Section 6.3). Rates move every asset — except when growth or earnings move them the other way (Section 5.2, Section 8.2). FOMC days are bigger than ordinary days — by a ratio of about 1.14 on the mean, with the direction entirely unknown in advance (Section 7.2). Each of these is a true statement about *tendencies over long horizons*, and each is nearly worthless as a *timing* tool over the horizons on which trades actually live.

There are four reasons macro fails at timing, and they are worth naming so the failure is not mistaken for bad luck. First, *long and variable lags*: the effect of a macro force arrives months or years later, and the delay itself varies from cycle to cycle, so even a correct call about direction gives no usable date (Section 5.3). Second, *anticipation*: markets price the *expected* path

continuously, so by the time a macro fact is obvious it is already in the price, and only the *surprise* moves anything (Section 7.1). Third, *reflexivity and regime change*: the relationships themselves shift — the same inflation number that hurts stocks in a hawkish regime helps them in a dovish one, and a rule fitted to one decade breaks in the next. Fourth, *non-stationarity of the data*: there are only a handful of business cycles in the reliable record, so any claim of the form "this has happened N times" rests on a sample far too small to trust the way one would trust a coin flipped a thousand times — a caution the macro-regimes paper and the quantitative papers make central.

The correct use of macro, then, is modest and real. It tells you what *kind* of environment you are trading in — whether growth is accelerating or slowing, whether inflation is a live threat, whether the central bank is a tailwind or a headwind, whether the day holds a scheduled shock. It sets the *odds* and the *backdrop*; it sizes your caution; it tells you when to expect the ground to move. It does not tell you what happens next hour, and any framework that claims it does should be distrusted precisely in proportion to its confidence. Written to learn, the honest conclusion of this paper is the one a good macro education always reaches: the tide is knowable; the waves are not; and confusing the two is the most expensive mistake in the subject.

11 Glossary and reading ladder

11.1 Glossary

Terms are listed roughly in order of first use, grouped by theme. Each definition is meant to stand alone.

Growth and the cycle. Macroeconomics — the study of the economy as a whole (growth, inflation, employment, rates) rather than of individual firms. **Gross Domestic Product (GDP)** — the total market value of goods and services produced in an economy in a period. **Real** — adjusted for inflation, i.e. measuring quantity; opposite of **nominal** (in current-price dollars). **Annualized** — a period's growth rate expressed as the yearly rate it would imply if sustained. **Business cycle** — the irregular wave of expansion and contraction in economic activity. **Expansion** — the phase of rising output and employment. **Peak** — the top of the cycle, the last month of expansion. **Contraction** — the phase of falling output. **Recession** — a significant, broad decline in activity lasting more than a few months. **Trough** — the bottom of the cycle, the last month of contraction. **NBER** — the National Bureau of Economic Research, whose committee officially dates U.S. recessions. **Industrial production** — the physical output of factories, mines and utilities; a coincident indicator. **Coincident indicator** — a statistic that turns at the same time as the cycle. **Lagging indicator** — one that turns after the cycle (e.g. unemployment). **Leading indicator** — one that tends to turn before the cycle (e.g. the yield-

curve slope). **Soft landing** — a slowdown that cools inflation without causing a recession. **Stagflation** — the uncomfortable combination of high inflation and stagnant growth.

Inflation. **Inflation** — the rate at which the general price level rises. **Deflation** — a falling general price level (negative inflation). **Disinflation** — a *slowing* of inflation that is still positive. **Price index** — a number tracking the cost of a fixed basket over time. **Consumer Price Index (CPI)** — the best-known U.S. price index, from the Bureau of Labor Statistics. **PCE price index** — the Personal Consumption Expenditures price index, the Fed's preferred gauge, from the Bureau of Economic Analysis. **Headline inflation** — the all-items rate. **Core inflation** — inflation excluding volatile food and energy, a cleaner trend read. **Year-over-year** — compared with the same month twelve months earlier. **Seasonally adjusted** — with the regular within-year pattern removed. **Base effect** — a change in a year-over-year rate caused by the *earlier* month's value rather than the recent one. **Demand-pull** — inflation from demand outrunning supply. **Cost-push** — inflation from rising input costs. **Monetary inflation** — sustained inflation from the money supply growing faster than output. **Transitory** — the label (later regretted) for the initially assumed temporary nature of the 2021 inflation.

Rates and central banks. **Interest rate** — the price of borrowing money, in percent per year. **Policy rate** — the short-term rate a central bank controls. **Federal funds rate** — the U.S. overnight interbank rate; the Fed's policy lever. **Effective federal funds rate** — the actual traded rate within the target range. **Target range** — the band the FOMC sets for the funds rate. **Basis point** — one hundredth of one percent. **Central bank** — the public institution managing a currency and monetary system. **Federal Reserve (Fed)** — the U.S. central bank. **European Central Bank (ECB)** — the euro area's central bank. **FOMC** — the Federal Open Market Committee, which sets U.S. rates. **Mandate** — a central bank's legal objectives. **Dual mandate** — the Fed's twin goals of maximum employment and stable prices. **Reaction function** — the systematic way a central bank responds to incoming data. **Discounting** — converting a future amount to its lower present value using an interest rate. **Present value** — what a future cash flow is worth today. **Transmission mechanism** — the chain by which policy-rate changes reach the economy. **Forward guidance** — a central bank's communication about the likely future path of rates. **Tightening / easing** — raising rates or draining liquidity / lowering rates or adding liquidity.

The bond market and liquidity. **Yield** — the annual return a bond offers if held to maturity. **Government bond / Treasury** — a (near) default-free loan to the government. **Maturity** — the time until a bond repays. **Yield curve** — yields plotted against maturity. **Level** — how high yields are in general. **Slope** — whether long yields exceed short yields. **Inversion** — long yields *below* short yields, a classic recession warning. **Term premium** — the extra yield demanded for holding longer maturities. **Quantitative easing (QE)** — a central bank creating money to buy

bonds, raising liquidity and lowering long yields. **Quantitative tightening (QT)** — the reverse: letting bonds run off, draining liquidity. **Balance sheet** — the assets a central bank holds; the standard liquidity gauge. **Reserves** — banks' money held at the central bank. **Liquidity** — the quantity of money and near-money in the system and the ease of moving it into assets. **Financial conditions** — a broad composite (rates, spreads, the dollar, equity levels) summarizing how loose or tight the overall environment is.

Data releases and cross-asset. Data release — an official statistic published at a pre-announced time. **Non-farm payrolls (NFP)** — the monthly U.S. employment report. **PMI / ISM** — purchasing-managers' survey indexes; above 50 signals expansion. **Retail sales** — monthly measure of consumer spending. **Surprise** — the gap between a released number and the expected one; what markets actually trade. **Soft data** — survey-based statistics (fast, forward-looking). **Hard data** — measured activity statistics (slower, backward-looking). **Consensus / expectation** — the surveyed forecast against which a release is judged. **Credit spread** — the extra yield a corporate bond pays over a government bond. **Foreign exchange (FX)** — the market for currencies. **Carry** — earning a higher interest rate by holding a higher-yielding currency or asset. **Safe haven** — an asset (Treasuries, the dollar, the yen, gold) sought in times of stress. **Risk-on / risk-off** — the single dial by which risk assets rise together (on) or safe havens rise together (off).

11.2 Reading ladder

For the reader who wants to go deeper, in increasing order of difficulty:

1. **This paper.** The four master variables and how they reach markets, from zero.
2. **A principles-of-macroeconomics textbook** — Mankiw's or Blanchard's core chapters on GDP, inflation, the business cycle, and monetary policy. The authoritative, careful version of everything sketched here.
3. **Central-bank educational material** — the Federal Reserve's *The Fed Explained* and the ECB's *Explainers*, both free and written for the public; the clearest first-party account of what central banks actually do.
4. **BIS explanatory work** — short, rigorous pieces from the Bank for International Settlements on monetary policy and the financial cycle, a step up in depth and candor about what is *not* well understood.
5. ****The macro-regimes paper, *Macro Regimes, Fundamentals and the Reaction Function***** — turns the qualitative picture here into something computable: growth–inflation regimes, the Taylor-rule reaction function, and financial-conditions indices, all built from public data.
6. ****The sentiment-and-positioning paper, *Sentiment and Positioning***** — measurable sentiment, positioning data, and the contrarian claim tested honestly.

7. ****Dalio, *Principles for Navigating Big Debt Crises***** — a practitioner's template for long-run debt cycles; read as a *model* to think with and to argue against, not as settled fact (see the note in the references).

Evidence basis

Every substantive claim in this paper rests on one of three kinds of basis, distinguished here so a reader can tell at a glance which is which — **own reproducible computation** (recomputed from free data or a self-contained model, reproducible via the cited script), **literature** (peer-reviewed article or textbook, cited inline), and **practitioner consensus** (desk practice with no clean citation, treated as the weakest). Nothing here is called *validated* on external sources alone; the stronger word requires an own out-of-sample computation.

Claim / result	Basis
The four US master variables since 1970 and their extreme readings — the 2020 output collapse and rebound, the 1970s and post-2021 inflation peaks, the ~\$9tn balance-sheet peak (Figure 1)	Own reproducible computation — figures/fig_e1_master_vars.py
Industrial production turns down into recessions while the 10-year-minus-3-month spread inverted before four of the five recessions since 1982 (Figure 2)	Own reproducible computation — figures/fig_e1_cycle.py
The 10-year-minus-2-year curve inverted before every recession since 1976 with lead times of ~7 to ~25 months, and the 2022 inversion had no dated recession by end of sample (Figure 4)	Own reproducible computation — figures/fig_e1_yield_curve.py
The 2021–23 inflation episode and policy response — CPI first above 2% in March 2021, ~9.0% peak (SA) in June 2022, funds-rate peak 5.50% in July 2023, first cut September 2024 (Figure 3)	Own reproducible computation — figures/fig_e1_inflation_episode.py
Scheduled FOMC decision days carry a larger average absolute S&P 500 move than ordinary days (~0.80% vs ~0.70%, ratio ~1.14) (Figure 5)	Own reproducible computation — figures/fig_e1_event_days.py
The yield-curve slope is among the best single financial predictors of US recessions	Literature — (Estrella & Mishkin 1998)
A recession is a broad decline dated by the NBER from several monthly series and announced with a long lag	Literature — (NBER Business Cycle Dating)
Monetary policy acts with long and variable lags	Literature — (Blanchard 2021)
The term spread is a real but imperfect recession predictor relative to the financial cycle	Literature — (Borio, Drehmann & Xia 2018)
The balance-sheet-to-risk-asset ("liquidity is everything") link is far looser than the popular framing; the portfolio-balance channel is genuine but modest	Practitioner consensus — not independently verified
Cross-asset couplings (rates→equities, credit, FX, commodities) are directional tendencies that are loose and regime-dependent	Practitioner consensus — not independently verified

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- Dalio, R. (2018). *Principles for Navigating Big Debt Crises*. Bridgewater / principles.com. Freely downloadable: <https://www.principles.com/big-debt-crises>. Cited as a *model* of the long-run debt cycle to reason with critically, not as established macroeconomics; its template is influential and its causal claims are the author's own.
- Data sources for the figures: U.S. Bureau of Labor Statistics (Consumer Price Index); Board of Governors of the Federal Reserve System (H.15 selected interest rates, H.4.1 balance sheet, G.17 industrial production); U.S. Department of the Treasury and the Federal Reserve Bank of New York (par-yield curve and effective federal funds rate / target range); U.S. Bureau of Economic Analysis and the OECD (real GDP growth); the National Bureau of Economic Research (recession dates); and Yahoo Finance (daily S&P 500 closes). All are freely and publicly accessible.

All references above are publicly accessible: two are widely published textbooks, one is an open-access working paper (direct link given), one is an open-access BIS working paper, three are free

first-party educational or data pages from the NBER, the Federal Reserve and the ECB, and one (Dalio) is a freely downloadable practitioner work cited critically as a model. No proprietary, course, or trading-academy material is cited.